

Task:

- Build deterministic math rules or models for each of the Kinematics axes here.

The models should be a light weight to run on device or the computations shouldn't be hard to compute

Requirements:

- Very light weight
- Can be run on device
- Can work with mobile or wearable devices as inputs
- Compare with other existing.
- Doesn't need to have thousands of features
- Should be able to adjust for phone orientation
- Augment the data to handle different phone orientations

Goal:

- Make it lightweight
- Better accuracy
- Generalizes for different orientations

Use the following starter notebooks we made using open dataset, they are google colab notebooks and you should be able to get them right.

Locomotion, Activity state have been experimented -

- [Colab link](#),
- [Colab link 2](#)
- [Locomotion](#)

Also to understand more about the kinematics axis read the following

Kinematic Axis

The kinematic axis answers the question: what is the body doing in the physical world?

It is based mainly on the 3-axis accelerometer, which measures movement in the X, Y, and Z directions. This axis is directly inferred from motion sensor data. It also helps identify when movement may be affecting physiological signals, such as heart rate, so those signals can be interpreted more reliably.

1. Activity State(have a starter notebook)

Activity state describes the current type of physical activity the person is doing. It is inferred from the strength and frequency pattern of accelerometer movement.

This is a broad category, not a continuous score, because activities such as sitting, walking, running, cycling, and driving are different types of movement rather than different points on one scale.

Possible activity states include:

- Sedentary: seated or lying down with minimal movement
- Standing: upright and mostly still
- Walking: rhythmic walking pattern
- Running: stronger, faster movement pattern
- Cycling: repeated cadence pattern
- Vigorous: high-intensity, irregular movement
- Driving: passive movement, useful as a research category

2. Postural State

Postural state describes the body's orientation or position relative to gravity. It is inferred from the gravity direction captured by the accelerometer.

This is important because posture changes physiological signals. For example, heart rate is usually higher when standing than when lying down. Because of this, posture helps the system understand whether a heart rate change may be due to body position rather than stress, emotion, or exertion.

Possible postural states include:

- Lying: body is horizontal
- Sitting: upright but mostly still
- Standing: upright with small postural movements
- In motion: body orientation is actively changing

3. Locomotion State(Have a starter notebook)

Locomotion state describes whether the body is moving through physical space and how that movement is happening.

This is different from activity state. For example, someone in a car is moving through space but not physically exerting themselves. Someone running on a treadmill is physically active but not really changing location. This distinction helps interpret the same sensor signals in the correct context.

Possible locomotion states include:

- Stationary: not moving through space
- Walking: slow self-powered movement

- Running: fast self-powered movement
- In vehicle: passive transport with low physical effort
- Ascending: moving upward, such as stairs or an incline
- Descending: moving downward, such as stairs or a decline

4. Movement Regularity

Movement regularity describes how smooth, stable, and predictable the person's movement is during a time window.

High movement regularity means movement is rhythmic and consistent, such as steady walking or calm stillness. Low movement regularity means movement is more erratic, variable, or unpredictable, such as fidgeting, restlessness, or agitation.

This can help support interpretation of cognitive and affective state. For example, calm stillness may support focus, while irregular movement may provide evidence of anxiety, agitation, or discomfort.